

JFET Common-Source Amplifier – Debugging Procedure

Problem: In LTspice simulation, the output voltage across the load resistor R_L remains at 0 V even though a sinusoidal input is applied.

1. Check the DC Supply

Measure the voltage between +VDD and ground. The circuit shows $V_{DD} = +20$ V. If the supply is absent or incorrectly connected, the amplifier cannot operate. Check the power-source value, polarity, and ground node.

2. Check the Drain Resistor R_D

Verify that $R_D = 2.2$ k Ω and that it is connected between +20 V and the JFET drain. Measure the drain DC voltage. A drain voltage stuck at 0 V or +20 V indicates a possible bias, JFET, or connection problem.

3. Check JFET Connections

Confirm the JFET terminals are correctly connected: gate to the input/bias network, source to ground, and drain to R_D . Also verify that the selected LTspice JFET model has the correct pin order. An incorrect model or pin mapping can produce zero output.

4. Check Gate Bias Resistor R_G

$R_G = 100$ M Ω and should connect the gate to ground. The gate DC voltage should be approximately 0 V because the source is grounded. Check for an open connection or an incorrect resistor value.

5. Check Input Coupling Capacitor C_1

$C_1 = 0.001$ μ F. Verify that the input source is connected through C_1 to the JFET gate. Measure the AC signal at the gate. If the input is present at the source but absent at the gate, check C_1 and its wiring.

6. Check Output Coupling Capacitor C_2

$C_2 = 0.001$ μ F and connects the drain to R_L . Measure the AC voltage at the drain and then across R_L . If an AC signal exists at the drain but not across R_L , check C_2 and its connections.

7. Check Load Resistor R_L

Verify $R_L = 32$ k Ω . Its lower terminal must be connected to ground. A wrong value or floating load connection can cause an incorrect or zero measured output.

8. Check the JFET Model and Parameters

Make sure the JFET model used in LTspice is appropriate for the circuit and that its parameters correspond to the device specified in the question. Check the model's gate, drain, and source pin mapping.

Required Measurements

Test point	Expected observation
VDD to Ground	Approximately +20 V DC
Gate to Ground	Approximately 0 V DC
Source to Ground	0 V DC

Drain to Ground	Non-zero DC voltage
Gate AC voltage	Input sinusoid should be present
Drain AC voltage	Amplified/inverted sinusoid
Across RL	Amplified AC output

Recommended Debugging Order

1. Verify +20 V supply → **2.** Verify ground → **3.** Verify JFET pin connections → **4.** Measure gate/source/drain DC voltages → **5.** Verify C1 and input signal → **6.** Verify drain AC signal → **7.** Verify C2 and RL → **8.** Verify JFET model.

Final Debugging Conclusion

The zero output is most likely caused by a missing DC supply, incorrect JFET connections/model, incorrect gate bias, or a broken input/output coupling path. After the DC operating point is correct and the sinusoidal signal reaches the gate, an amplified and phase-inverted AC signal should appear at the drain and across RL.